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IEC 61499 Compliant Web-based IDE for Function Block Design

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List of Acronyms

CPPS	Cyber-Physical Production Systems
FBD	Function Block Oriented Programming
PLC	Programmable Logic Controller
RTE	Runtime Environment
IDE	Integrated Development Environment
ICS	Industrial Control Systems
SOA	Service-Oriented Architectures

Chapter 1

Introduction

1.1 Context

Cyber-Physical Production Systems (**CPPS**) are complex systems that integrate physical processes with digital technologies to optimise production processes. Several software engineering approaches and technologies are utilised in the development and operation of **CPPS**. One of such approaches is Function Block Oriented Programming (**FBD**). **FBD** is a graphical programming language used in industrial automation and control systems. **FBD** uses graphical function blocks to represent the logic and control of a system. The function blocks are connected in a network to form a control program. One of the benefits of **FBD** is that it provides a visual representation of the control logic, which can be easier to understand and debug than traditional textual programming languages. **FBD** is also modular, which means that function blocks can be reused in multiple programs, reducing development time and improving code maintainability. Also, **FBD** is often used in conjunction with Programmable Logic Controllers (**PLCs**) to control machinery and processes in manufacturing and other industrial applications. **FBD** programs can be created and edited using specialised software tools that allow for the creation and manipulation of function blocks.

Several methods and tools are used in **FBD**, including the IEC 61499 standard. As IEC 61499 supporting tools, there are Runtime Environments (**RTEs**) and Integrated Development Environments (**IDEs**). While **RTEs** enables the execution of Function Blocks compliant with the standard, **IDEs** are used to design, create, and deploy Function Block pipelines into the Edge layer of a **CPPS**.

These **IDEs** offer features and tools that facilitate the development, testing, and deployment of IEC 61499-compliant control systems. They provide graphical editors for creating function block diagrams, simulation capabilities, and additional functionalities to streamline the development process. However, despite being increasingly popular, there aren't many web-based **IDEs** specifically designed for IEC 61499.

1.2 Motivation

The evolution of **CPPS** has increased the demand for flexible and intelligent control solutions capable of integrating physical processes with digital technologies. **FBD** and the IEC 61499 standard offer a structured, modular approach to building such distributed automation systems. To fully leverage these benefits, development tools must evolve alongside industrial needs, supporting more accessible, collaborative, and scalable workflows.

1.3 Problem

Existing IEC 61499 development environments are predominantly desktop-based and lack modern capabilities such as remote access, real-time collaboration, and seamless integration with cloud or edge infrastructures. These limitations hinder efficient development, maintenance, and deployment of control applications within distributed **CPPS**. As a result, current tools fall short in supporting the flexibility, scalability, and interconnectedness required by modern industrial automation systems.

1.4 Research Questions

This research seeks to investigate how modern web technologies can be leveraged to enhance the development, deployment, and operation of IEC 61499-compliant systems. This study is guided by the following main research question:

How can web-based **IDEs** enhance the development, deployment, and collaboration processes for IEC 61499 applications in distributed **CPPS**, addressing the limitations of traditional desktop-based environments?

To further refine the scope, the following sub-questions are proposed:

- What architectural and technological requirements must be considered when designing a web-based **IDEs** for IEC 61499?
- In what ways do web-based **IDEs** improve collaboration, accessibility, and scalability compared to existing desktop tools?

Chapter 2

Literature Review

This chapter reviews prior work related to IEC 61499-based development, **CPPS**, and web-based **IDEs**. The goal is to identify how current tools support distributed automation and where existing gaps justify the development of a web-based solution.

2.1 Methodology

A systematic literature review approach was adopted, the objective was to collect, screen, and analyze studies addressing IEC 61499 development tools, **CPPS**, and web-based engineering environments.

2.1.1 Search and Selection

The search was conducted in *ScienceDirect*, *Scopus* and others using combinations of the following terms:

("IEC 61499" OR "FBD") AND ("web-based" OR "IDE") AND ("CPPS")

2.2 Related Work

Recent studies explore cloud-native automation and containerized function block execution [1, 2], yet most solutions remain constrained by traditional architectures. Emerging web-based IDE prototypes, such as early 4diac Web IDE developments [3], show promise in accessibility and scalability but lack maturity and full IEC 61499 compliance.

Chapter 3

State of the Art

This chapter provides a comprehensive analysis of the current landscape regarding development environments for **CPPS** and the IEC 61499 standard. It categorizes existing solutions into traditional desktop-based environments and emerging web-based engineering tools while identifying the key technological enablers that facilitate the migration of **FBD** to the web.

3.1 Development Ecosystem

Currently, the development ecosystem is dominated by desktop-based **IDEs**. These tools are typically monolithic applications installed locally on engineering workstations.

3.1.1 Open-Source

The most prominent open-source implementation is the Eclipse 4diac framework. It consists of the 4diac IDE (based on the Eclipse framework) and the 4diac FORTE (a run-time environment), although it can also use different **RTEs** like DINASORE.

This IDE is very feature rich at the cost of significant system resources and increased complexity which make it more difficult to use. It relies on the heavyweight Eclipse Rich Client Platform.

3.1.2 Commercial

The commercial side is dominated by EcoStruxure Automation Expert, ISaGRAF Workbench and other monolithic desktop based environments, usually with a built-in proprietary **RTE**. This makes them often very complete production ready tools, but they are not very flexible and not the best fit for many needs.

3.2 Evolution of Web-Based Engineering Tools

The software engineering domain has seen a massive shift from desktop **IDEs** to cloud-native environments. This trend is now influencing the domain of Industrial Control Systems (**ICS**).

Recent studies have proposed Service-Oriented Architectures (SOA) to integrate IEC 61499 with cloud layers, yet the engineering tools themselves remain largely desktop-bound.

3.2.1 Web-Based IDEs in Software Development

In general software development, tools like Visual Studio Code have started offering web-based counterparts to their desktop applications in search of a more flexible user experience.

3.3 Gap Analysis

Current IEC 61499 tools face significant limitations that hinder modern CPPS development:

- **Platform Dependency:** Existing solutions are monolithic desktop applications, lacking the accessibility and portability of web-based environments.
- **Resource Intensity:** Current options are heavyweight and complex, creating a high barrier to entry compared to modern agile tooling.
- **The Web Gap:** Despite industry trends, there is currently no mature, lightweight web interface for FBD, leaving ICS engineering behind the curve of cloud-native software development.

References

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